

04_yolo_detection_counting

July 5, 2026

0.0.1 Installing the YOLO Framework

```
[ ]: !pip install ultralytics -q
```

```
41.8/41.8 kB
1.9 MB/s eta 0:00:00
1.3/1.3 MB
20.0 MB/s eta 0:00:00
```

```
[ ]: import os
import cv2
import matplotlib.pyplot as plt
from ultralytics import YOLO
```

Creating new Ultralytics Settings v0.0.6 file
View Ultralytics Settings with 'yolo settings' or at
'/root/.config/Ultralytics/settings.json'
Update Settings with 'yolo settings key=value', i.e. 'yolo settings
runs_dir=path/to/dir'. For help see
<https://docs.ultralytics.com/quickstart/#ultralytics-settings>.

```
[ ]: !unzip -q banana_dataset.zip -d dataset/
```

```
[ ]: base_dir = '/content/dataset'
train_dir = os.path.join(base_dir, 'train')
val_dir = os.path.join(base_dir, 'valid')
```

1 YOLOv8 CLASSIFICATION (Ripeness Grading)

```
[ ]: # Load the YOLOv8 nano classification model
model_cls = YOLO('yolov8m-cls.pt')
```

Downloading
<https://github.com/ultralytics/assets/releases/download/v8.4.0/yolov8m-cls.pt> to
'yolov8m-cls.pt': 100% 32.7MB 127.8MB/s 0.3s

1.0.1 Training the model

```
[ ]: print("Starting YOLO Classification Training...")

# Create a data.yaml file for YOLO training
# This defines the dataset structure for YOLOv8 classification
with open(os.path.join(base_dir, 'data.yaml'), 'w') as f:
    f.write(f"path: {base_dir}\n") # Dataset root directory
    f.write("train: train/\n") # Train images relative to path
    f.write("val: valid/\n") # Val images relative to path
    f.write("nc: 3\n") # Number of classes
    f.write("names: ['overripe', 'ripe', 'unripe']\n") # Class names

results_cls = model_cls.train(
    data=base_dir, # Pass the path to the dataset root directory
    epochs=15,      # 10 epochs is usually sufficient for YOLOv
    ↪fine-tuning
    imgsz=416,      # Using the 416x416 size specified in the project
    project='yolo_models', # Folder where results will be saved
    name='banana_cls' # Subfolder for this specific run
)

yolo_cls_path = '/content/yolo_models/banana_cls/weights/best.pt'
```

Starting YOLO Classification Training..

Ultralytics 8.4.87 Python-3.12.13 torch-2.11.0+cu128 CUDA:0 (Tesla T4, 14913MiB)

engine/trainer: agnostic_nms=False, amp=True, angle=1.0, augment=False, auto_augment=randaugument, batch=16, bgr=0.0, box=7.5, cache=False, cfg=None, classes=None, close_mosaic=10, cls=0.5, cls_pw=0.0, compile=False, conf=None, copy_paste=0.0, copy_paste_mode=flip, cos_lr=False, cutmix=0.0, data=/content/dataset, degrees=0.0, deterministic=True, device=, dfl=1.5, dis=6.0, distill_model=None, dnn=False, dropout=0.0, dynamic=False, embed=None, end2end=None, epochs=15, erasing=0.4, exist_ok=False, fliplr=0.5, flipud=0.0, format=torchscript, fraction=1.0, freeze=None, hsv_h=0.015, hsv_s=0.7, hsv_v=0.4, imgsz=416, iou=0.7, keras=False, kobj=1.0, line_width=None, lr0=0.01, lrf=0.01, mask_ratio=4, max_det=300, mixup=0.0, mode=train, model=yolov8m-cls.pt, momentum=0.937, mosaic=1.0, multi_scale=0.0, name=banana_cls-2, nbs=64, nms=False, opset=None, optimize=False, optimizer=auto, overlap_mask=True, patience=100, perspective=0.0, plots=True, pose=12.0, pretrained=True, profile=False, project=yolo_models, quantize=None, rect=False, resume=False, retina_masks=False, rle=1.0, save=True, save_conf=False, save_crop=False, save_dir=/content/runs/classify/yolo_models/banana_cls-2, save_frames=False, save_json=False, save_period=-1, save_txt=False, scale=0.5, seed=0, shear=0.0, show=False, show_boxes=True, show_conf=True, show_labels=True, simplify=True, single_cls=False, source=None, split=val, stream_buffer=False, task=classify, time=None, tracker=tracktrack.yaml, translate=0.1, val=True, verbose=True,

```

vid_stride=1, visualize=False, warmup_bias_lr=0.1, warmup_epochs=3.0,
warmup_momentum=0.8, weight_decay=0.0005, workers=8, workspace=None
train: /content/dataset/train... found 11793 images in 4 classes
val: /content/dataset/valid... found 1123 images in 4 classes
test: /content/dataset/test... found 562 images in 4 classes
Overriding model.yaml nc=1000 with nc=4

      from  n    params  module
arguments
  0          -1  1      1392  ultralytics.nn.modules.conv.Conv
[3, 48, 3, 2]
  1          -1  1     41664  ultralytics.nn.modules.conv.Conv
[48, 96, 3, 2]
  2          -1  2     111360  ultralytics.nn.modules.block.C2f
[96, 96, 2, True]
  3          -1  1     166272  ultralytics.nn.modules.conv.Conv
[96, 192, 3, 2]
  4          -1  4     813312  ultralytics.nn.modules.block.C2f
[192, 192, 4, True]
  5          -1  1     664320  ultralytics.nn.modules.conv.Conv
[192, 384, 3, 2]
  6          -1  4    3248640  ultralytics.nn.modules.block.C2f
[384, 384, 4, True]
  7          -1  1    2655744  ultralytics.nn.modules.conv.Conv
[384, 768, 3, 2]
  8          -1  2    7084032  ultralytics.nn.modules.block.C2f
[768, 768, 2, True]
  9          -1  1     990724  ultralytics.nn.modules.head.Classify
[768, 4]
YOLOv8m-cls summary: 80 layers, 15,777,460 parameters, 15,777,460 gradients,
41.9 GFLOPs
Transferred 228/230 items from pretrained weights
AMP: running Automatic Mixed Precision (AMP) checks...
Downloading
https://github.com/ultralytics/assets/releases/download/v8.4.0/yolo26n.pt to
'yolo26n.pt': 100%          5.3MB 69.8MB/s 0.1s
AMP: checks passed
train: Fast image access (ping: 0.0±0.0 ms, read: 817.3±183.8
MB/s, size: 23.5 KB)
train: Scanning /content/dataset/train... 11793 images, 0
corrupt: 100%          11793/11793 5.3Kit/s 2.2s
train: New cache created: /content/dataset/train.cache
val: Fast image access (ping: 0.0±0.0 ms, read: 555.7±261.7 MB/s,
size: 12.7 KB)
val: Scanning /content/dataset/valid... 1123 images, 0 corrupt:
100%          1123/1123 3.8Kit/s 0.3s
val: New cache created: /content/dataset/valid.cache
optimizer: 'optimizer=auto' found, ignoring 'lr0=0.01' and

```

'momentum=0.937' and determining best 'optimizer', 'lr0' and 'momentum' automatically...

optimizer: AdamW(lr=0.00125, momentum=0.9) with parameter groups 38 weight(decay=0.0), 39 weight(decay=0.0005), 39 bias(decay=0.0)

Image sizes 416 train, 416 val

Using 2 dataloader workers

Logging results to /content/runs/classify/yolo_models/banana_cls-2

Starting training for 15 epochs...

Epoch	GPU_mem	loss	Instances	Size
Downloading https://ultralytics.com/assets/Arial.ttf to				
'/root/.config/Ultralytics/Arial.ttf': 100%				755.1KB 13.2MB/s 0.1s
1/15	2.01G	0.4081	1	416: 100%

738/738 3.7it/s 3:18

classes	top1_acc	top5_acc: 100%	36/36 6.3it/s
---------	----------	----------------	---------------

5.7s

all	0.965	1
-----	-------	---

Epoch	GPU_mem	loss	Instances	Size
2/15	2.45G	0.1757	1	416: 100%
738/738 3.9it/s 3:10				
classes	top1_acc	top5_acc: 100%	36/36 6.3it/s	

5.7s

all	0.965	1
-----	-------	---

Epoch	GPU_mem	loss	Instances	Size
3/15	2.45G	0.1741	1	416: 100%
738/738 3.9it/s 3:10				
classes	top1_acc	top5_acc: 100%	36/36 6.5it/s	

5.5s

all	0.968	1
-----	-------	---

Epoch	GPU_mem	loss	Instances	Size
4/15	2.45G	0.1512	1	416: 100%
738/738 3.9it/s 3:10				
classes	top1_acc	top5_acc: 100%	36/36 6.3it/s	

5.7s

all	0.968	1
-----	-------	---

Epoch	GPU_mem	loss	Instances	Size
5/15	2.45G	0.1059	1	416: 100%
738/738 3.9it/s 3:10				
classes	top1_acc	top5_acc: 100%	36/36 6.3it/s	

5.7s

all	0.975	1
-----	-------	---

Epoch	GPU_mem	loss	Instances	Size
6/15	2.45G	0.09135	1	416: 100%

738/738	3.9it/s	3:09			
	classes	top1_acc	top5_acc: 100%	36/36	6.6it/s
5.4s	all	0.973	1		
	Epoch	GPU_mem	loss	Instances	Size
	7/15	2.45G	0.07299	1	416: 100%
738/738	4.0it/s	3:07			
	classes	top1_acc	top5_acc: 100%	36/36	5.3it/s
6.8s	all	0.98	1		
	Epoch	GPU_mem	loss	Instances	Size
	8/15	2.45G	0.0569	1	416: 100%
738/738	3.9it/s	3:08			
	classes	top1_acc	top5_acc: 100%	36/36	6.1it/s
5.9s	all	0.984	1		
	Epoch	GPU_mem	loss	Instances	Size
	9/15	2.45G	0.05491	1	416: 100%
738/738	3.9it/s	3:07			
	classes	top1_acc	top5_acc: 100%	36/36	6.2it/s
5.8s	all	0.987	1		
	Epoch	GPU_mem	loss	Instances	Size
	10/15	2.45G	0.05183	1	416: 100%
738/738	3.9it/s	3:07			
	classes	top1_acc	top5_acc: 100%	36/36	5.4it/s
6.7s	all	0.984	1		
	Epoch	GPU_mem	loss	Instances	Size
	11/15	2.45G	0.04159	1	416: 100%
738/738	3.9it/s	3:09			
	classes	top1_acc	top5_acc: 100%	36/36	6.5it/s
5.5s	all	0.983	1		
	Epoch	GPU_mem	loss	Instances	Size
	12/15	2.45G	0.03454	1	416: 100%
738/738	3.9it/s	3:07			
	classes	top1_acc	top5_acc: 100%	36/36	5.3it/s
6.8s	all	0.989	1		
	Epoch	GPU_mem	loss	Instances	Size

```

13/15      2.45G      0.02841      1      416: 100%
738/738 4.0it/s 3:07
      classes      top1_acc      top5_acc: 100%      36/36 6.5it/s
5.5s
      all      0.986      1

```

```

Epoch      GPU_mem      loss      Instances      Size
14/15      2.45G      0.03354      1      416: 100%
738/738 4.0it/s 3:05
      classes      top1_acc      top5_acc: 100%      36/36 5.4it/s
6.6s
      all      0.988      1

```

```

Epoch      GPU_mem      loss      Instances      Size
15/15      2.45G      0.01999      1      416: 100%
738/738 3.9it/s 3:09
      classes      top1_acc      top5_acc: 100%      36/36 6.6it/s
5.4s
      all      0.986      1

```

15 epochs completed in 0.830 hours.

Optimizer stripped from

/content/runs/classify/yolo_models/banana_cls-2/weights/last.pt, 31.7MB

Optimizer stripped from

/content/runs/classify/yolo_models/banana_cls-2/weights/best.pt, 31.7MB

Validating /content/runs/classify/yolo_models/banana_cls-2/weights/best.pt...

Ultralytics 8.4.87 Python-3.12.13 torch-2.11.0+cu128 CUDA:0 (Tesla T4, 14913MiB)

YOLOv8m-cls summary (fused): 42 layers, 15,767,780 parameters, 0 gradients, 41.6 GFLOPs

train: /content/dataset/train... found 11793 images in 4 classes

val: /content/dataset/valid... found 1123 images in 4 classes

test: /content/dataset/test... found 562 images in 4 classes

```

      classes      top1_acc      top5_acc: 100%      36/36 6.8it/s
5.3s
      all      0.989      1

```

Speed: 0.3ms preprocess, 2.5ms inference, 0.0ms loss, 0.0ms postprocess per image

Results saved to /content/runs/classify/yolo_models/banana_cls-2

2 YOLOv8 DETECTION AND COUNTING

```

[ ]: print("\n--- YOLOv8 Object Detection and Counting ---")

# 1. Load the pre-trained YOLOv8 nano detection model

```

```

model_det = YOLO('yolov8n.pt') # Changed from classification model to detection,
    ↳model

# 2. Saving a copy of this pre-trained model to match required project
    ↳deliverables
model_det.save('yolo_detect_best.pt')
print("Detection model saved as 'yolo_detect_best.pt'")

# 3. Creating a function to detect, count, and visualize bananas
def detect_and_count_bananas(image_path):
    """
    Takes an image path, uses YOLOv8 to detect bananas (COCO Class 46),
    draws bounding boxes, and prints the total count.
    """
    # Run inference. 'classes=[46]' forces the model to ONLY look for bananas.
    results = model_det.predict(source=image_path, classes=[46], conf=0.25)

    # Extract the result object
    result = results[0]

    # The number of boxes detected equals the number of bananas
    banana_count = len(result.boxes)

    # Plot the results (draws the bounding boxes on the image)
    annotated_img = result.plot()

    # Convert BGR (OpenCV format) to RGB (Matplotlib format) for correct colors
    annotated_img_rgb = cv2.cvtColor(annotated_img, cv2.COLOR_BGR2RGB)

    # Display the image and the count
    plt.figure(figsize=(8, 8))
    plt.imshow(annotated_img_rgb)
    plt.axis('off')
    plt.title(f"Bananas Detected: {banana_count}", fontsize=16,
    ↳fontweight='bold', color='blue')
    ↳fontweight='bold', color='blue')
    plt.show()

    return banana_count

```

```

--- YOLOv8 Object Detection and Counting ---
Detection model saved as 'yolo_detect_best.pt'

```

3 Testing the Counter

```
[ ]: # Let's grab a random image from your test set to see it in action
# Adjust this path if you want to test a specific image!
test_image_folder = os.path.join(base_dir, 'test/overripe')
sample_images = os.listdir(test_image_folder)

if len(sample_images) > 0:
    sample_image_path = os.path.join(test_image_folder, sample_images[0])
    print(f"Testing on image: {sample_images[0]}")

    # Run our function!
    count = detect_and_count_bananas(sample_image_path)
    print(f"Final Count for this image: {count}")
else:
    print("Could not find a test image in the specified directory.")
```

Testing on image: musa-acuminata-overripe-9c147c05-1d0a-11ec-96f3-d8c4975e38aa_jpg.rf.bf5019eb3a0c8f03701e09a11444d259.jpg

image 1/1 /content/dataset/test/overripe/musa-acuminata-overripe-9c147c05-1d0a-11ec-96f3-d8c4975e38aa_jpg.rf.bf5019eb3a0c8f03701e09a11444d259.jpg: 640x640 1 banana, 7.8ms
Speed: 3.1ms preprocess, 7.8ms inference, 1.5ms postprocess per image at shape (1, 3, 640, 640)

Bananas Detected: 1



Final Count for this image: 1